

AI Emergency Operations & Resource Orchestration System

Bhote Valley Emergency Simulation

IBM Bob AI Innovation Hackathon 2026

Problem

Disaster response operations are overwhelmed by fragmented, uncertain, and conflicting information from satellites, field teams, hospitals, and sensors. Without evidence fusion, accessibility analysis, and resource optimization, critical decisions are delayed.

Solution

An evidence-aware operational decision engine that fuses multi-source observations with uncertainty quantification, detects conflicting reports, computes graph-based route accessibility, and optimizes resource allocation — all backed by deterministic algorithms and IBM Bob MCP integration.

Architecture

React Dashboard + IBM Bob → FastAPI + MCP Server → Simulation Engine (Evidence Fusion | Graph Router | Resource Optimizer | Simulation) → SQLite

IBM Technology Integration

14 MCP tools via IBM Bob providing natural-language operational Q&A; backed by the deterministic evidence and routing engines. All answers come from actual backend state — no hardcoded responses.

Impact

64-test suite covering evidence fusion, conflict detection, routing, optimization, and simulation. Benchmark demonstrates measurable improvement over nearest-resource baseline strategy.